

Course 08: App Deployment

Comprehensive Learning Guide & Practical Prompting Playbook

MODULE OVERVIEW

Bridge development and production. Master prompts for configuring CI/CD pipelines, Dockerizing applications, configuring cloud hosting (Railway, AWS, Vercel), setup environment security, and automated monitoring.

KEY LEARNING TOPICS

- Containerization with Docker & Multi-Stage Builds
- CI/CD Pipeline Design (GitHub Actions, GitLab CI)
- Cloud Platform Deployment (Railway, Render, Vercel, AWS)
- Environment Security, Secrets Management, & Health Monitoring

PROMPTING STRATEGY & ARCHITECTURE

Provide exact deployment targets, OS environments, build scripts, and security requirements to generate production-grade DevOps configurations.

ROLE

You are a Senior DevOps and Cloud Infrastructure Engineer.

TASK

Create a production-ready `Dockerfile` and a GitHub Actions workflow (`.github/workflows/deploy.yml`) for a Node.js Express backend deploying to Railway.

CONTEXT

- Application: Node.js 20, npm, uses Express and Mongoose.
- Needs multi-stage build to minimize image size.
- Environment variables required: `PORT`, `MONGO\_URI`, `JWT\_SECRET`.

INSTRUCTIONS

1. Write optimized multi-stage `Dockerfile` (build phase vs production phase).
2. Create `.dockerignore` contents.
3. Write GitHub Actions YAML file that runs tests on Pull Request and deploys to Railway on main branch push.
4. List key deployment verification steps.

BEST PRACTICES & PRACTICAL TIPS

- Never expose sensitive credentials or keys in prompts; use placeholders like ``.
- Ask AI to generate post-deployment verification scripts and health check endpoints.
- Prompt AI to review configurations for security vulnerabilities (e.g., non-root Docker execution).